

**BOMBALA, Australia**WMO: **949290**Lat: **37.0017S**Lon: **149.2336E**Elev: **2498**StdP: **13.42**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 25.9       | 28.0 | 19.6                          | 16.1 | 38.8 | 23.7 | 19.6 | 36.6 | 32.2                  | 46.7 | 29.2 | 47.2 | 3.9                   | 340  | 0.609 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 24.6                   | 88.4            | 63.6 | 83.7 | 62.5 | 79.6 | 61.9 | 67.7                | 77.5 | 66.0 | 75.8 | 64.6 | 74.4 | 13.9                 | 300  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.2                            | 102.2 | 70.3 | 63.2 | 95.3 | 68.4 | 61.6 | 89.8 | 67.0 | 33.6          | 77.2 | 32.1 | 76.1 | 31.0 | 74.5 | 86.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 28.4          | 24.6 | 21.3 | DB                         | 21.5 | 95.7               | 1.9 | 4.1                                                | 20.2 | 98.7       | 19.1 | 101.1      | 18.0 | 103.4      | 16.6 | 106.4 | (4) |
| (5)               |      |      | WB                         | 21.2 | 70.8               | 1.9 | 2.0                                                | 19.9 | 72.2       | 18.8 | 73.4       | 17.7 | 74.5       | 16.4 | 75.9  | (5) |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.1   | 64.0  | 62.6  | 58.7  | 52.3  | 46.6  | 42.5  | 41.2  | 43.0  | 47.6  | 51.5  | 56.1  | 59.8  | (6)  |  |
|                                                                        | DBStd       | 9.52   | 7.09  | 6.40  | 6.05  | 5.08  | 4.56  | 3.74  | 3.70  | 4.23  | 5.22  | 6.22  | 6.69  | 6.78  | (7)  |  |
|                                                                        | HDD50       | 1073   | 1     | 1     | 4     | 32    | 126   | 226   | 273   | 221   | 108   | 57    | 18    | 6     | (8)  |  |
|                                                                        | HDD65       | 4879   | 107   | 112   | 209   | 382   | 569   | 674   | 736   | 681   | 523   | 420   | 276   | 190   | (9)  |  |
|                                                                        | CDD50       | 1843   | 434   | 354   | 275   | 101   | 22    | 2     | 1     | 5     | 36    | 103   | 201   | 309   | (10) |  |
|                                                                        | CDD65       | 174    | 75    | 45    | 15    | 0     | 0     | 0     | 0     | 0     | 1     | 9     | 29    |       | (11) |  |
|                                                                        | CDH74       | 2457   | 966   | 535   | 209   | 12    | 0     | 0     | 0     | 0     | 3     | 54    | 215   | 463   | (12) |  |
|                                                                        | CDH80       | 857    | 399   | 191   | 50    | 1     | 0     | 0     | 0     | 0     | 0     | 5     | 55    | 156   | (13) |  |
| (14) Wind                                                              | WSAvg       | 8.8    | 8.9   | 8.2   | 7.8   | 7.3   | 8.3   | 8.1   | 9.1   | 9.8   | 10.1  | 9.8   | 9.2   | 9.0   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 26.0   | 2.4   | 2.2   | 2.6   | 1.9   | 1.7   | 2.0   | 1.9   | 1.5   | 1.8   | 2.2   | 3.1   | 2.4   | (15) |  |
|                                                                        | PrecMax     | 40.6   | 5.7   | 15.4  | 7.0   | 6.5   | 7.2   | 10.9  | 7.0   | 4.6   | 6.5   | 6.9   | 11.0  | 6.3   | (16) |  |
|                                                                        | PrecMin     | 14.1   | 0.1   | 0.0   | 0.1   | 0.0   | 0.2   | 0.2   | 0.0   | 0.4   | 0.1   | 0.3   | 0.1   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 7.2    | 1.6   | 2.4   | 1.9   | 1.6   | 1.5   | 2.1   | 1.8   | 1.1   | 1.2   | 1.5   | 2.2   | 1.5   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 95.9  | 92.4  | 86.1  | 75.3  | 66.6  | 59.1  | 58.5  | 64.8  | 71.7  | 79.9  | 86.4  | 91.3  | (19) |  |
|                                                                        |             | MCWB   | 65.2  | 64.3  | 62.6  | 58.1  | 54.4  | 49.1  | 47.9  | 49.6  | 52.9  | 56.8  | 60.5  | 62.4  | (20) |  |
|                                                                        | 2%          | DB     | 89.4  | 85.5  | 79.5  | 70.1  | 62.5  | 55.7  | 54.9  | 59.4  | 66.8  | 74.4  | 79.9  | 84.4  | (21) |  |
|                                                                        |             | MCWB   | 64.4  | 63.3  | 61.2  | 56.2  | 52.8  | 48.5  | 46.2  | 47.7  | 51.8  | 56.3  | 59.8  | 61.6  | (22) |  |
|                                                                        | 5%          | DB     | 84.1  | 80.6  | 74.9  | 66.4  | 59.2  | 53.0  | 52.1  | 55.9  | 63.1  | 69.3  | 75.1  | 79.2  | (23) |  |
|                                                                        |             | MCWB   | 64.3  | 63.0  | 61.2  | 55.1  | 51.4  | 47.1  | 44.7  | 46.3  | 51.1  | 55.1  | 58.9  | 61.1  | (24) |  |
|                                                                        | 10%         | DB     | 78.8  | 75.6  | 70.3  | 62.7  | 56.1  | 50.4  | 49.5  | 52.8  | 59.1  | 64.8  | 70.3  | 74.1  | (25) |  |
|                                                                        |             | MCWB   | 63.3  | 62.8  | 60.0  | 54.4  | 49.6  | 45.8  | 43.8  | 44.8  | 49.4  | 53.8  | 58.2  | 60.4  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.1  | 69.6  | 67.5  | 61.2  | 57.9  | 53.5  | 50.7  | 52.2  | 56.9  | 62.0  | 65.8  | 67.2  | (27) |  |
|                                                                        |             | MCDB   | 82.1  | 80.7  | 74.0  | 68.7  | 61.7  | 56.2  | 54.5  | 59.8  | 64.5  | 69.9  | 75.1  | 77.8  | (28) |  |
|                                                                        | 2%          | WB     | 68.1  | 67.4  | 65.0  | 58.7  | 54.5  | 50.0  | 47.9  | 49.5  | 54.3  | 59.2  | 63.1  | 64.9  | (29) |  |
|                                                                        |             | MCDB   | 79.5  | 76.1  | 72.8  | 65.3  | 59.8  | 53.4  | 51.8  | 56.4  | 62.7  | 66.7  | 72.5  | 75.9  | (30) |  |
|                                                                        | 5%          | WB     | 66.2  | 65.6  | 62.9  | 57.1  | 52.4  | 48.2  | 45.9  | 47.5  | 52.2  | 57.0  | 61.1  | 63.1  | (31) |  |
|                                                                        |             | MCDB   | 77.1  | 74.6  | 70.3  | 63.1  | 57.5  | 51.3  | 50.3  | 53.8  | 60.2  | 65.7  | 70.6  | 73.7  | (32) |  |
|                                                                        | 10%         | WB     | 64.6  | 64.0  | 61.1  | 55.3  | 50.3  | 46.5  | 44.3  | 45.6  | 50.2  | 54.9  | 59.0  | 61.5  | (33) |  |
|                                                                        |             | MCDB   | 75.3  | 72.6  | 68.1  | 60.9  | 54.9  | 49.4  | 48.9  | 51.3  | 57.4  | 63.6  | 68.3  | 71.2  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 24.6  | 22.0  | 20.9  | 20.1  | 18.4  | 16.2  | 17.1  | 19.1  | 20.3  | 21.9  | 22.6  | 23.3  | (35) |  |
|                                                                        |             | MCDBR  | 34.1  | 31.1  | 29.0  | 26.7  | 24.0  | 21.0  | 21.7  | 25.0  | 27.7  | 31.1  | 32.6  | 32.8  | (36) |  |
|                                                                        |             | MCWBR  | 12.7  | 12.0  | 12.7  | 13.9  | 14.9  | 14.1  | 13.9  | 14.7  | 14.7  | 15.4  | 14.7  | 13.3  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 28.2  | 25.0  | 23.0  | 21.8  | 20.2  | 17.2  | 18.5  | 21.8  | 24.0  | 25.2  | 26.5  | 26.7  | (38) |  |
|                                                                        |             | MCWBR  | 12.8  | 11.7  | 11.9  | 12.5  | 13.4  | 12.7  | 13.1  | 13.8  | 14.0  | 14.5  | 14.6  | 13.1  | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.386 | 0.348 | 0.313 | 0.301 | 0.281 | 0.271 | 0.264 | 0.274 | 0.296 | 0.312 | 0.324 | 0.333 | (40) |  |
|                                                                        | taud        |        | 2.355 | 2.503 | 2.596 | 2.599 | 2.619 | 2.649 | 2.680 | 2.639 | 2.565 | 2.543 | 2.543 | 2.525 | (41) |  |
|                                                                        | Ebn at Noon |        | 300   | 305   | 305   | 290   | 279   | 273   | 284   | 296   | 305   | 314   | 318   | 319   | (42) |  |
|                                                                        | Edh at Noon |        | 42    | 34    | 29    | 25    | 21    | 19    | 20    | 24    | 29    | 33    | 34    | 35    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2099  | 1796  | 1519  | 1141  | 843   | 659   | 761   | 1033  | 1421  | 1765  | 1963  | 2144  | (44) |  |
|                                                                        | RadStd      |        | 160   | 142   | 120   | 88    | 41    | 46    | 54    | 57    | 102   | 110   | 144   | 147   | (45) |  |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (27 neighbors) | +0.72 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -140  | +207  | +90   |

Nomenclature: See separate page